ChatGPT

Deep Research Report: Section 2 Literature Map and Research-Gap Audit

Executive research verdict

The five-part structure already encoded in `sections/02_literature.tex` is well chosen: real-time forecasting; mixed-frequency/bridge nowcasting; factors and shrinkage; combinations and instability; and a research-gap subsection written only after the preceding literature has been established. The present Introduction also formulates H1 as a **fixed-model comparison across information stages** and H2 as a **stage-dependent policy versus a fixed policy**, with model/policy selection confined to pre-evaluation data.

The central finding of this literature audit is consequential:

> **The paper should not claim that within-quarter stage dependence, changing model rankings, adaptive model selection, real-time vintages, or forecasting multiple macroeconomic variables are individually novel. All have substantial prior art.**

There is strong evidence that model performance changes as information accumulates within a quarter. Aastveit et al. update U.S. GDP density nowcasts after every new release and explicitly report that the ranking of model classes changes during the quarter; their combination also outperforms a simple model-selection strategy. [1] Maurin and Drechsel construct six forecasts during a quarter using different information sets, indicators, individual equations and pooling weights. [2] More recently, Heinrich and Reif compare several model classes over three within-quarter information sets using genuine ALFRED data and show that both absolute accuracy and relative model performance depend on how much intra-quarter information has arrived. [3]

Adaptive model choice is also established. Giacomini and White develop conditional predictive-ability tests and a decision rule that uses current information to choose the forecast expected to perform best. [4] Koop and Korobilis allow the entire forecasting model to change over time through dynamic model averaging and dynamic model selection, and implement the exercise with real-time U.S. inflation data. [5] Kuzin, Marcellino and Schumacher compare sequential model selection with pooling in large mixed-frequency nowcasting systems. [6]

The promising research gap is narrower and, in my assessment, substantially more defensible:

> **Among the close papers reviewed, I did not find an exact implementation of a pre-specified, nested pseudo-real-time experiment that first isolates the effect of information arrival by holding the forecasting rule fixed, then isolates the incremental effect of a release-stage-conditioned model policy at the same information stage, freezes that policy using a separate development sample before final evaluation, repeats the comparison under historically available and latest-vintage data, and applies the architecture across GDP, inflation and labour-market targets.**

That should be the paper's positioning. It is a contribution in **experimental decomposition, policy discipline and breadth of evaluation**, not a claim that stage-specific forecasting itself is new. This conclusion follows particularly from the contrast between the information-flow literature, which holds or largely holds the forecasting architecture fixed while information changes, and the adaptive-selection literature, which changes models conditional on state or past performance but generally does not set up the same-stage H1/H2 counterfactual proposed here. [7]

The novelty audit can therefore be summarised as follows.

Potential claim	Verdict	Safe positioning
Forecast accuracy changes as releases arrive	**Already well established**	Treat as motivation and H1 to be quantified across your targets, not as novelty. [8]
Model rankings change within the quarter	**Already established**	Cite Aastveit et al. directly and acknowledge it. [1]
Forecast specifications can differ by release stage	**Already established**	Maurin–Drechsel is particularly dangerous to any broad novelty claim. [2]
Adaptive/dynamic model selection	**Already established**	Position H2 against Giacomini–White, Koop–Korobilis and Kuzin et al. [9]
Stage analysis with genuine real-time data	**Already established**	Aastveit et al., Liebermann and Heinrich–Reif are clear precedents. [10]
Forecasting GDP, inflation and labour variables in one broad framework	**Exists in related forms**	Do not claim that cross-domain forecasting itself is new. Heinrich–Reif jointly study GDP, CPI and unemployment, and related mixed-frequency systems are explicitly multivariate. [11]
Separate information arrival from model-policy choice	**Closest gap**	Claim that your design *explicitly separates* the two contrasts; avoid “first” unless a later exhaustive citation check supports it.
Freeze a stage-policy on development data and test it on untouched observations	**Not found in the close comparator set**	Present as an anti-overfitting design feature and part of the contribution, not as a universal claim about all forecasting literature. [12]
Real-time versus latest-vintage counterfactual under otherwise identical origins/rules	**Literature exists on the vintage issue; your controlled implementation is useful but not foundationally novel**	Anchor H3 in Croushore/Stark, Koenig et al. and subsequent real-time work. [13]

Literature corpus and intellectual ancestry

The search screened **42 distinct scholarly works** closely enough to determine their relationship to the proposed design. The corpus spans the foundational real-time-data literature, release-flow nowcasting, bridge/MIDAS and mixed-frequency models, factor and shrinkage methods, forecast combinations, instability and adaptive model selection, and recent multi-target real-time systems. The objective was not merely to collect citations but to locate papers that could invalidate an overly broad novelty claim.

The **real-time and vintage-data family** should be anchored by Croushore and Stark's real-time dataset work; Stark and Croushore's explicit comparison of real-time with latest-available forecasting data; Koenig, Dolmas and Piger's analysis of alternative ways to estimate forecasting equations with real-time data; Croushore's later survey; Faust and Wright's large real-time dataset comparison; Liebermann's daily real-time GDP nowcasting exercise; Golinelli and Parigi's adaptive real-time work; and the recent Čapek et al. assessment of how revisions change DSGE forecast performance. Stark and Croushore are especially useful for H3 because they show that forecast accuracy measures can be vintage-sensitive and may appear deceptively favourable when latest-available data replace historically available observations. Koenig et al. likewise demonstrate that the way real-time observations enter estimation materially affects forecasting performance. [14]

The **real-time information-flow family** centres on Giannone, Reichlin and Small; Bańbura and Rünstler; Bańbura, Giannone, Modugno and Reichlin; Aastveit et al.; Evans; Liebermann; Maurin and Drechsel; Schorfheide and Song; Carriero, Clark and Marcellino; and Beber, Brandt and Luisi. Giannone et al. are foundational because they formalise updating nowcasts of output and inflation as new releases arrive and interpret release-level changes through economic “news”. [15] Bańbura and co-authors place that exercise in the broader real-time data-flow setting, while Aastveit et al. explicitly update a competing-model system release by release. [16] Evans demonstrates how scheduled announcements can be converted into daily real-time estimates of macroeconomic conditions, potentially for GDP, inflation or unemployment. [17] Beber, Brandt and Luisi similarly extract high-frequency information from releases concerning output, employment, inflation and sentiment. [18]

The **mixed-frequency and bridge/MIDAS family** includes Schumacher and Breitung; Schumacher's comparison of MIDAS and bridge equations; Schorfheide and Song's mixed-frequency VAR; Carriero et al.'s Bayesian mixed-frequency model; Maurin and Drechsel's bridge-equation pooling system; recent time-varying MIDAS work by Chan, Poon and Zhu; the group-shrinkage MIDAS approach of Kohns and Potjagailo; Heinrich and Reif's time-varying mixed-frequency VARs; and Moreira's 2025 Bank of England SC-MIDAS architecture. Schorfheide and Song explicitly document the value of information that becomes available during the quarter in a real-time MF-VAR. [19] Carriero et al. allow the model to expand as additional within-quarter observations become available, making the connection between the ragged edge and changing forecast specification explicit. [20] Moreira's recent Bank of England implementation is especially relevant for modern practice because it dynamically changes the relative contribution of hard and soft information through the release cycle rather than treating all stages identically. [21]

The **factor/high-dimensional/shrinkage family** includes Stock and Watson's diffusion-index work; De Mol, Giannone and Reichlin's large-predictor Bayesian shrinkage analysis; Bańbura, Giannone and Reichlin's large Bayesian VAR; Schumacher and Breitung's factor nowcasting; Bok et al.'s New York Fed nowcasting system; Kant, Pick and de Winter's machine-learning comparison; Franjic's predictor-preselection work; and recent evidence from Shen and Xiu on ridge versus lasso under weak signals. Stock and Watson establish the canonical logic of summarising very large predictor panels with a small number of estimated factors. [22] De Mol et al. show that strongly regularised Bayesian regression can provide forecasts closely related to principal-component forecasts in large panels. [23] Bańbura, Giannone and Reichlin extend the shrinkage logic to large VARs. [24] These are the natural intellectual ancestors for the paper's factor and ridge/shrinkage benchmarks; the contribution should lie in how those methods are evaluated across stages, not in their use per se.

The **forecast-combination, instability and adaptation family** includes the Bates–Granger tradition as synthesised by Timmermann; Stock and Watson's instability literature; Deutsch, Granger and Teräsvirta on changing combination weights; Giacomini and White on conditional predictive ability and forecast selection; Koop and Korobilis on dynamic model averaging/selection; Kuzin, Marcellino and Schumacher on pooling versus selection; Aastveit et al.'s two generations of density-combination work; and the more recent stage-aware combination architectures at central banks. Timmermann's survey provides the standard rationale for combination under model uncertainty and explains why simple combinations can perform surprisingly well relative to estimated optimal weights. [25] Kuzin et al. find substantial time variation in the performance of individual nowcasting specifications and comparatively stable performance from pooling. [6] Aastveit et al. demonstrate the same practical issue within the release calendar itself. [1]

Finally, the search deliberately examined **cross-domain systems**, because a claim that no one jointly studies activity, prices and labour conditions would now be untenable. Giannone et al. explicitly consider real-time output and inflation. [15] Aruoba and Diebold develop real-time monitoring of real activity and inflation jointly. [26] Most importantly for the proposed contribution, Heinrich and Reif conduct a genuinely real-time ALFRED-based forecast experiment covering U.S. GDP, industrial production, CPI and unemployment within one mixed-frequency VAR architecture and evaluate multiple within-quarter information sets. [3] Trinh's 2026 mixed-frequency BVAR similarly generates forecasts for CPI, GDP, the cash rate and unemployment. [27] The defensible contribution is therefore the **common stage-policy experiment over a broader eight-target set**, rather than “the first joint GDP/ inflation/labour framework”.

For planning purposes, the 42-work screening set is:

Family	Works screened most closely
Real-time/vintages	Croushore & Stark (2001); Stark & Croushore (2002); Koenig, Dolmas & Piger (2003); Croushore (2011); Faust & Wright; Golinelli & Parigi; Liebermann (2014); Čapek et al. (2025). [14]
Information flow	Giannone, Reichlin & Small (2008); Bańbura & Rünstler (2011); Bańbura et al. (2013); Evans (2005); Beber, Brandt & Luisi; Maurin & Drechsel; Aastveit et al. (2014); Aastveit et al. (2018). [28]
Mixed-frequency/ bridge	Schumacher & Breitung; Schumacher; Schorfheide & Song (2015); Schorfheide & Song (2021); Carriero, Clark & Marcellino; Heinrich & Reif; Chan, Poon & Zhu; Kohns & Potjagailo; Moreira. [29]
Factors/shrinkage/ ML	Stock & Watson; De Mol, Giannone & Reichlin; Bańbura, Giannone & Reichlin; Bok et al.; Kant, Pick & de Winter; Franjic; Shen & Xiu. [30]
Combination/ instability/ adaptation	Timmermann; Stock & Watson (instability); Stock & Watson (forecast combinations/predictive instability); Deutsch, Granger & Teräsvirta; Giacomini & White; Koop & Korobilis; Kuzin, Marcellino & Schumacher; Aastveit et al. (2014, 2018). [31]

Family	Works screened most closely
Cross-domain boundary cases	Aruoba & Diebold; Evans; Heinrich & Reif; Trinh; Giannone, Reichlin & Small. [32]

This is enough depth for Section 2. There is no need to force all 42 into the manuscript; about **25–30 carefully chosen citations** should be sufficient for the actual literature section, with the rest serving as novelty checks and robustness references.

Claim-source matrix for the five subsections

The matrix below is the key output to use when drafting. "Establishes" deliberately states **only what the cited work supports**; the final column indicates the paragraph in which the claim belongs.

Claim usable in Section 2	Best source(s)	What the source actually establishes	Use
Real-time macroeconomic forecasting operates on an incomplete, asynchronously updated information set.	Giannone, Reichlin & Small; Bańbura et al.	Macroeconomic releases arrive sequentially and nowcasts are revised as new observations/news become available. [33]	§2.1
Real-time information flow can be examined release by release rather than only through conventional forecast horizons.	Giannone et al.; Liebermann	Giannone et al. measure contributions of releases; Liebermann tracks daily real-time GDP-nowcast performance through the data flow. [34]	§2.1
Real-time and latest-available datasets are not interchangeable for forecasting exercises.	Stark & Croushore	Forecast-model choices and measured accuracy can be vintage-sensitive; latest data can make errors appear deceptively smaller. [35]	§2.1
The estimation method itself must respect historical vintages.	Koenig, Dolmas & Piger	Alternative methods of constructing estimation samples from real-time data give different outcomes; their preferred real-time procedure improves forecasting performance. [36]	§2.1
Revised data can change conclusions about large-data forecasting.	Faust & Wright	Real-time large-dataset forecasts can differ from results obtained using ex-post revised predictor data. [37]	§2.1
Data revisions remain consequential in modern forecasting models.	Čapek et al.	Real-time versus revised observations affect predictive performance of DSGE models, with heterogeneous effects across variables and economies. [38]	§2.1

Claim usable in Section 2	Best source(s)	What the source actually establishes	Use
Bridge equations are a natural response to low-frequency targets and timely high-frequency indicators.	Maurin & Drechsel	Individual bridge equations based on monthly indicators are used to nowcast quarterly GDP and are pooled across models. [2]	§2.2
Publication lags change which predictors are useful at different forecast origins.	Giannone et al.; Bańbura/Rünstler tradition	Timeliness and economic content are distinct aspects of the value of a release; real-time usefulness depends on publication timing. [15]	§2.2
MIDAS and bridge equations solve related mixed-frequency problems but impose different aggregation/forecasting structures.	Schumacher	Direct comparison of MIDAS and bridge-equation approaches identifies their different treatments of high-frequency information. [39]	§2.2
Mixed-frequency VARs allow monthly information to improve forecasts of quarterly variables without first reducing everything to quarterly frequency.	Schorfheide & Song	Real-time MF-VAR forecasts explicitly exploit intra-quarter observations and improve on quarterly-frequency alternatives. [19]	§2.2
Ragged-edge models may effectively change dimension as information arrives.	Carriero, Clark & Marcellino	Their Bayesian mixed-frequency specification expands as additional within-quarter indicators become observable. [20]	§2.2
Stage-specific specifications are not themselves novel.	Maurin & Drechsel	Six within-quarter forecasts use different information sets, indicators, equations and combination weights. [2]	§2.2 / §2.5
Large predictor panels can be reduced to a few common factors.	Stock & Watson	Approximate factor/diffusion-index methods forecast macroeconomic series from very large predictor panels. [22]	§2.3
Shrinkage offers an alternative to factor compression in high dimensions.	De Mol, Giannone & Reichlin	Strongly regularised Bayesian regressions can forecast comparably to principal-component methods in large panels. [23]	§2.3
Large VARs can be made feasible through shrinkage.	Bańbura, Giannone & Reichlin	Shrinkage scaled to system dimension makes large Bayesian VARs useful for forecasting and structural analysis. [24]	§2.3

Claim usable in Section 2	Best source(s)	What the source actually establishes	Use
Predictor preselection is an additional route to controlling high-dimensional nowcasting models.	Franjic	Elastic-net preselection can improve mixed-frequency dynamic-factor nowcasting in the studied application. 40	§2.3
Ridge deserves explicit treatment when many predictors carry weak signals.	Shen & Xiu	Their theoretical and empirical evidence favours ridge relative to lasso in low signal-to-noise/high-dimensional settings. 41	§2.3
Model ranking can depend on the point in the release calendar.	Aastveit et al.	Model-class rankings change during the quarter as new U.S. GDP information arrives. 1	§2.4
This stage dependence is still visible in recent machine-learning comparisons.	Kant, Pick & de Winter	Different methods lead at different pseudo-real-time nowcast horizons/stages for Dutch GDP. 42	§2.4
It also occurs in genuine real-time multi-variable systems.	Heinrich & Reif	Model accuracy differs across I1, I2 and I3 information sets for GDP, CPI, unemployment and industrial production using ALFRED data. 3	§2.4
Forecast combinations are a central defence against model uncertainty.	Timmermann	Combination can outperform individual candidates when model selection is uncertain, although estimated weighting rules face estimation error. 25	§2.4
Pooling can be more robust than selecting one nowcasting specification.	Kuzin et al.	Individual performance varies substantially; selection results depend on the rule; pooling is comparatively stable. 6	§2.4
Combinations can be especially useful when model rankings change within the release calendar.	Aastveit et al.	Their density combination performs well across stages and exceeds a simple model-selection strategy. 1	§2.4
Adaptive forecast selection conditioned on current information is established econometric methodology.	Giacomini & White	Conditional predictive-ability framework plus a two-step forecast-selection rule based on information available at the decision date. 4	§2.4
Entire forecasting models can be allowed to change recursively over time.	Koop & Korobilis	Dynamic model averaging/ selection lets the predictor set and model identity evolve in a real-time inflation exercise. 5	§2.4

Claim usable in Section 2	Best source(s)	What the source actually establishes	Use
Stage-dependent weighting is already used in contemporary policy nowcasting.	Moreira	Bank of England SC-MIDAS dynamically exploits hard versus soft data through the release cycle. [21]	§2.4 / §2.5
A common architecture spanning activity, inflation and labour variables is not unprecedented.	Heinrich & Reif; Trinh	The former evaluates GDP, CPI, industrial production and unemployment with one family of real-time MF models; the latter forecasts GDP, CPI and unemployment, among other variables, with an MF-BVAR. [11]	§2.5
The defensible gap is therefore not "stage dependence" in isolation.	Synthesis of Aastveit; Maurin–Drechsel; Giacomini–White; Koop–Korobilis	Prior work separately covers stage-varying rankings/ specifications and conditional/ dynamic model selection. [43]	§2.5
The proposed H1/H2 design imposes an additional identification discipline on forecast comparisons.	Current MacroPulse design plus preceding literature	MacroPulse explicitly defines H1 with the model fixed and H2 against a same-problem fixed policy, whereas the closest papers do not present that exact nested contrast. [44]	§2.5

The matrix also identifies an important rhetorical rule: **citations to Aastveit et al. and Maurin–Drechsel belong in the research-gap paragraph itself, not only earlier in the review.** They are the strongest evidence against an inflated novelty claim and therefore make the eventual narrower contribution statement more credible. [45]

Research-gap audit against the dangerous novelty questions

The answer to **"Has anyone already compared model performance across within-quarter release stages?" is unambiguously yes.** Aastveit et al. update U.S. GDP nowcasts after every release and find changing model-class rankings. [1] Liebermann tracks a fully real-time factor model through the daily data flow and finds nowcast precision increases as information is released. [46] Heinrich and Reif construct three within-quarter information sets and report model-specific RMSEs for each. [3] Kant, Pick and de Winter likewise obtain different winning approaches across pseudo-real-time nowcast horizons. [42] **Therefore the paper must not say that comparing model performance by release stage is novel.**

The answer to **"Has anyone implemented stage-specific model selection or switching rather than simply updating one model?" is also broadly yes, although the exact mechanism varies.** Maurin and Drechsel produce six intra-quarter forecasts with stage-dependent sets of indicators, bridge equations and pooling weights. [2] Aastveit et al. compare a release-by-release multi-model combination to a simple model-selection strategy and show that the ranking of model classes changes during the quarter. [1] Kuzin et al. implement sequential model selection in a nowcasting context, though their conditioning variable is not a fixed release-stage policy of the MacroPulse form. [6] More generally, Giacomini–White conditional selection and Koop–Korobilis dynamic selection show that changing the forecasting rule according to available information or evolving model probabilities is established methodology. [47] **Therefore "we introduce stage-dependent model selection" is too broad.**

For **"Have stage-dependent policies been evaluated with genuinely real-time vintages?"**, there are close precedents. Aastveit et al. use U.S. real-time data and update competing model classes throughout the quarter. [1] Liebermann explicitly constructs historical data vintages sufficient to reproduce information available on each historical date. [46] Heinrich and Reif use ALFRED real-time data and multiple within-quarter information sets. [3] Koop and Korobilis combine genuine real-time observations with dynamic model adaptation, albeit at quarterly forecast dates rather than a within-period release-stage policy. [48] **The remaining opening is the conjunction of real-time vintages with a frozen, development-sample-selected mapping $s \mapsto m_s$, evaluated against a same-stage fixed-model counterfactual.**

For **"Is there prior work jointly covering GDP, inflation and labour-market nowcasting under one architecture?"**, the answer is sufficiently close to yes that no exclusivity claim is safe. Heinrich and Reif jointly forecast GDP, industrial production, CPI and unemployment using one family of mixed-frequency models with genuine real-time data. [3] Trinh's 2026 MF-BVAR includes GDP, CPI and unemployment in a common system. [27] Evans describes a framework capable of producing real-time estimates of GDP, inflation or unemployment, although his empirical focus is GDP. [17] Giannone et al. jointly address output and inflation. [15] What remains distinctive is **the same explicit H1–H3 policy-comparison architecture applied to a much wider target vector—GDP, four inflation concepts and three labour-market outcomes—each with its own release calendar.**

For **"How common is development-sample selection followed by untouched evaluation for adaptive forecasting policies?"**, the literature reviewed here does **not** support saying that this is standard practice. Close papers more commonly update models, probabilities, weights, hyperparameters or selection criteria recursively as the forecast origin moves forward. Koop–Korobilis update model probabilities in real time; Giacomini–White's decision rule uses information available at the selection date; Kuzin et al. use sequential model-selection procedures; and Kant et al. tune models within evolving estimation samples. [49] This review did not identify a close nowcasting paper that selects a deterministic **target-by-release-stage policy on a dedicated development period, freezes that mapping before an untouched final evaluation period, and uses that final period solely for policy assessment**. That is potentially a strong design feature. The safe wording is "unlike the closest approaches reviewed here" rather than "unlike the literature as a whole".

For **"What does the literature say about real-time versus latest-vintage evaluation?"**, the answer is unequivocal: using revised data can alter model specification, forecast errors and ranking conclusions. Stark and Croushore directly warn that RMSE and MAE calculated with latest data can be deceptively low and that real-time benchmark comparisons should use information actually available historically. [35] Koenig et al. show that even the construction of the estimation dataset matters. [36] Faust and Wright explicitly motivate large real-time datasets partly because standard large-predictor evidence often relies on ex-post revised observations unavailable to actual forecasters. [37] The issue remains active: Čapek et al. find meaningful effects of revisions on modern macro-model forecast performance. [38] This provides strong ancestry for H3 rather than a novelty claim.

For **"Is the proposed H1/H2 separation already present somewhere?"**, the answer is the most nuanced. The **ingredients** plainly exist, but the **explicit two-contrast experimental decomposition was not located in this search**. Giannone et al. supply the clearest ancestor of H1 by examining what new releases add as the information set expands. [15] Schorfheide–Song and Heinrich–Reif likewise assess the value of intra-quarter information within specified model families. [50] Giacomini–White and Koop–Korobilis supply the intellectual ancestry of H2 by allowing forecast choice to respond to information or evolving model performance. [47] Aastveit et al. come closest to joining the two ideas because they observe model-ranking changes throughout the real-time release cycle and compare combination against selection. [1] Maurin–Drechsel also allow equations and weights to differ across information dates. [2] But these papers do not formulate the particular pair of controlled loss contrasts now specified by MacroPulse. That distinction is the strongest candidate for the research gap.

Why the H1–H2 decomposition is genuinely useful

The current Introduction already contains the right basic idea, but Section 2 should make the contrast conceptually precise. Let

$$\ell_{t,s}(m) = L\left(e_{t,s}^{(m)}\right)$$

denote forecast loss when rule m is applied to information set $\mathcal{I}_{t,s}$.

The **information-arrival contrast** is

$$\Delta^{I}_{t,s}(m) = \ell_{t,s+1}(m) - \ell_{t,s}(m).$$

Only the information stage changes. The forecasting rule is held fixed:

$$m \quad \text{fixed}, \qquad \mathcal{I}_{t,s} \longrightarrow \mathcal{I}_{t,s+1}.$$

This is the logical descendant of the real-time news/data-flow literature. Giannone et al. analyse how new information changes nowcasts; Schorfheide and Song document forecast improvement from within-quarter information; Liebermann tracks precision along the real-time data flow; and Heinrich–Reif compare the same model classes at multiple within-quarter information sets. [51]

The **stage-policy contrast** should instead be defined at the **same stage and therefore the same information set**:

$$\Delta^{P}_{t,s} = \ell_{t,s}(m_s^*) - \ell_{t,s}\left(m^F\right),$$

where m_s^* is the model selected in advance for stage s and m^F is the target-specific fixed comparator. Here,

$$\mathcal{I}_{t,s} \quad \text{fixed}, \qquad m^F \longrightarrow m_s^*.$$

That is what makes H2 conceptually different from merely observing that later nowcasts are better. Conditional and dynamic forecast selection already have deep antecedents in Giacomini–White and Koop–Korobilis, but those papers condition selection on current information/evolving performance rather than using release stage as the fixed policy state in the proposed nested experiment. [47]

The entire design can be represented as a simple two-dimensional comparison:

	Fixed rule m^F	Pre-specified stage rule m_s^*
Information stage s	$\ell_{t,s}(m^F)$	$\ell_{t,s}(m_s^*)$

	Fixed rule m^F	Pre-specified stage rule m_s^*
Information stage $s+1$	$\ell_{t,s+1}(m^F)$	$\ell_{t,s+1}(m_{s+1}^*)$

H1 moves **vertically in the first column**:

$$\ell_{t,s+1}(m^F) - \ell_{t,s}(m^F),$$

whereas H2 moves **horizontally within a row**:

$$\ell_{t,s}(m_s^*) - \ell_{t,s}(m^F).$$

That simple organisation is not just rhetorical. It prevents a common interpretation problem in nowcasting comparisons. When a later forecast uses a richer information set and a different specification simultaneously—as can occur in stage-varying bridge systems—the resulting improvement cannot by itself tell us whether the gain arose because more data became available or because the forecasting rule changed. Maurin–Drechsel's six-date architecture is an important prior-art example of information sets, equations and weights changing together. [2] Aastveit et al. similarly demonstrate that information content and relative model performance both evolve through the quarter. [1] The MacroPulse design can add value by **measuring those dimensions separately before examining them jointly**.

I recommend tightening the eventual methodology in three ways.

First, "fixed model" in H1 should mean a fixed **forecasting rule, specification logic and tuning protocol**, not necessarily a literal regression with an identical rectangular design matrix. Ragged-edge models naturally receive different observed entries at different stages. Carriero et al., for example, explicitly accommodate changing within-quarter availability. [20] The important condition is that m does not change *because the researcher selected a different model at stage* $s+1$.

Second, the H2 mapping

$$\mathcal{P}(k,s) = m_{k,s}^*$$

should be estimated entirely on the policy-development sample and then frozen before the final evaluation period. That is stronger than recursively asking at every evaluation origin which candidate has recently performed best. Dynamic and conditional selection procedures commonly update through time, as in Giacomini–White and Koop–Korobilis. [47] A frozen policy makes H2 a cleaner test of whether **release stage contains stable enough information about relative model performance to support an ex-ante policy**.

Third, H3 must keep the release mask fixed when creating the latest-vintage counterfactual. In other words,

$$\mathcal{I}_{t,s}^{RT} \quad \text{versus} \quad \mathcal{I}_{t,s}^{LV}$$

should differ in the **values of revisable observations**, not in whether data that had not yet been released historically are allowed to appear. Otherwise H3 would conflate revision effects with the H1 information-arrival mechanism. The real-time literature's concern is precisely that historically unavailable revised values can contaminate retrospective forecasting exercises. [52] Your existing ALFRED validation already establishes empirically that GDP growth calculated from different historical snapshots can differ materially, providing project-specific motivation for this controlled counterfactual. filecite turn0file0

Exact citation-to-paragraph map for `sections/02_literature.tex`

The current file has exactly the five subsections this research supports. I recommend **seventeen substantive paragraphs** in total. This is enough to establish ancestry and the gap without turning Section 2 into a catalogue.

Paragraph	Main claim to establish	Primary citations	What not to claim
§2.1 P1	Define the real-time forecasting problem: asynchronously released observations generate evolving/ ragged information sets; nowcasts are updated as news arrives.	Giannone, Reichlin & Small; Bańbura et al.; Liebermann. [53]	Do not yet discuss model selection.
§2.1 P2	A historically feasible information set is different from a rectangular dataset constructed later; data vintages affect estimation and forecast evaluation.	Croushore/Stark; Stark/Croushore; Koenig–Dolmas–Piger. [54]	Do not imply every revision materially changes every forecast.
§2.1 P3	The vintage problem is empirically important and persists in large-data and modern-model settings.	Faust & Wright; Čapek et al. [55]	Do not claim real-time data always worsen accuracy.
§2.1 P4	Establish the paper's conceptual distinction between release-stage information and vintage values. Release stage controls *what is observable*; vintage controls *which historical value is observed*.	Synthesis anchored in Giannone et al., Stark–Croushore and Koenig et al. [56]	Present this as organising synthesis, not a theorem attributable to one source.
§2.2 P1	Bridge equations use timely high-frequency indicators to forecast low-frequency targets; publication calendars make stage-specific information sets natural.	Maurin & Drechsel; Bańbura/Rünstler lineage. [57]	Do not describe bridge equations as novel.
§2.2 P2	MIDAS and mixed-frequency models provide alternatives to bridge-style temporal aggregation and can use intra-period observations directly.	Schumacher; Schorfheide & Song; Carriero et al. [58]	Do not imply one family dominates universally.
§2.2 P3	The value of a model depends on stage: real-time performance changes as releases arrive.	Aastveit et al.; Liebermann; Heinrich & Reif; Kant et al. [59]	Explicitly concede that stage-by-stage performance analysis already exists.

Paragraph	Main claim to establish	Primary citations	What not to claim
§2.2 P4	Some previous systems allow indicators, equations or weights themselves to vary through the release calendar.	Maurin & Drechsel; Moreira. [60]	This paragraph is essential to prevent a false claim that stage-specific forecasting is new.
§2.3 P1	Large information sets motivated factor/diffusion-index methods.	Stock & Watson; Schumacher/ Breitung if needed. [61]	Avoid a general history of factor analysis.
§2.3 P2	Shrinkage is an alternative/ complement to dimensionality reduction; large Bayesian regression/VARs can exploit many predictors while regularising coefficients.	De Mol et al.; Bańbura–Giannone–Reichlin. [62]	Do not equate every form of shrinkage with ridge.
§2.3 P3	Modern preselection, ridge and ML methods enlarge the candidate-model set but create tuning/model-selection uncertainty.	Franjic; Shen & Xiu; Kant et al. [63]	Keep this focused on why these are candidate model classes for MacroPulse.
§2.4 P1	Forecast combinations are a standard response to model uncertainty and unstable rankings.	Timmermann; Kuzin et al. [64]	Do not assert combinations always dominate.
§2.4 P2	Predictive relationships and relative performance can change over time; conditional forecast selection can exploit observable state information.	Stock–Watson instability; Giacomini & White. [65]	Distinguish statistical possibility from guaranteed exploitable gains.
§2.4 P3	Dynamic model averaging/selection allows model identity to evolve recursively and has been implemented with real-time macro data.	Koop & Korobilis. [5]	Therefore "adaptive model selection" cannot be the novelty claim.
§2.4 P4	The nowcasting literature connects model uncertainty directly to the release cycle: model rankings vary across stages and combinations often provide robust performance.	Aastveit et al. 2014/2018; Kuzin et al.; Moreira. [66]	End by setting up the need to distinguish changing information from changing policy.
§2.5 P1	Concede the closest prior art: stage-wise information evaluation, stage-varying specifications, adaptive selection, real-time vintages and cross-domain systems all exist separately or in close combinations.	Aastveit; Maurin–Drechsel; Giacomini–White; Koop–Korobilis; Heinrich–Reif. [67]	**Do not use "no previous study considers…" here.**

Paragraph	Main claim to establish	Primary citations	What not to claim
§2.5 P2	State the actual gap: the reviewed work does not jointly implement the nested H1/H2 decomposition, a development-sample-frozen stage policy, identical same-stage comparators, genuine-vintage/latest-vintage counterfactuals, and the proposed broad target suite.	Synthesis of preceding sources plus MacroPulse hypotheses. [68]	Say “we did not identify in the closest literature” or “the literature has not generally combined”; avoid “this is the first”.
§2.5 P3	Position the contribution as a disciplined empirical comparison of forecasting policies rather than a new model: H1 isolates information arrival; H2 isolates stage policy; H3 isolates vintage effects.	MacroPulse hypotheses and real-time/adaptive ancestors. [69]	Do not promise superiority before final results.

That structure should give Section 2 a clear argument rather than a chronological literature review:

$$\text{real-time feasibility} \rightarrow \text{mixed-frequency information flow} \rightarrow \text{candidate model families} \rightarrow \text{model uncertainty/ad}$$

It also follows the comment already present in your source file to organise the literature “around the paper's problem” rather than chronologically.

Positioning decisions and citation priorities

For the final §2.5, the **strongest defensible contribution language** is approximately this proposition, which should later be converted into manuscript prose rather than copied verbatim:

> Existing nowcasting research shows that information arriving within the target period can improve forecasts, that relative model performance can vary over the release cycle, and that combinations or adaptive selection can mitigate model uncertainty. Real-time-data research separately establishes that forecast evaluation can be sensitive to historical vintages. The present analysis does not treat these ingredients as individually novel. Instead, it organises them in a common pseudo-real-time experiment designed to distinguish the loss change associated with an expanding information set from the incremental loss change associated with a pre-specified stage-dependent model policy, while holding the information stage fixed for the latter comparison. The policy is selected before the final evaluation sample, and the same experiment is repeated under genuine historical and latest-vintage data across heterogeneous GDP, inflation and labour-market targets.

Every element of the first two sentences is backed by close prior work. [70] The last two sentences describe the conjunction that this search did not find in a close comparator.

The corresponding **unsafe formulations** are: “the first stage-dependent nowcasting framework”; “previous studies hold the model fixed as information arrives”; “no prior study allows model choice to depend on release stage”; “real-time nowcasting has focused only on GDP”; “no previous study jointly forecasts GDP, inflation and labour-market conditions”; and “existing research does not distinguish information arrival from model adaptation". Aastveit et al., Maurin–Drechsel, Koop–Korobilis and Heinrich–Reif provide direct counterexamples to significant parts of those statements. [71]

A slightly stronger sentence is defensible if carefully qualified:

> **"To our knowledge, the closest literature does not jointly impose the two controlled comparisons used here: a fixed-rule cross-stage contrast for information arrival and a same-stage fixed-versus-stage-policy contrast, with the stage policy frozen prior to final evaluation."**

The phrase "to our knowledge" is necessary because a literature search cannot logically prove the non-existence of a paper. More importantly, the sentence defines the alleged gap narrowly enough that the closest predecessors do not contradict it. Aastveit et al. compare models through the release cycle but do not formulate this nested fixed-rule/same-stage decomposition; Maurin–Drechsel vary both information and forecasting architecture across their six dates; Giacomini–White develop conditional selection but not a release-stage policy of this form; Koop–Korobilis dynamically update model choice over calendar time rather than freezing a release-stage mapping before an untouched evaluation sample. [43]

The **highest-priority citation set** for the actual Section 2 should therefore be roughly 27 works rather than all 42 screened:

Subsection	Highest-priority works
§2.1	Croushore & Stark; Stark & Croushore; Koenig–Dolmas–Piger; Croushore review; Giannone–Reichlin–Small; Liebermann; Faust–Wright; Čapek et al. [72]
§2.2	Bańbura/Rünstler; Maurin–Drechsel; Schumacher; Schorfheide–Song; Carriero–Clark–Marcellino; Aastveit et al.; Heinrich–Reif; Moreira. [73]
§2.3	Stock–Watson diffusion indexes; De Mol–Giannone–Reichlin; Bańbura–Giannone–Reichlin; Bok et al.; Kant–Pick–de Winter; Franjic; Shen–Xiu. [30]
§2.4	Timmermann; Stock–Watson instability; Giacomini–White; Koop–Korobilis; Kuzin–Marcellino–Schumacher; Aastveit et al. 2014/2018. [74]
§2.5	Aastveit; Maurin–Drechsel; Giacomini–White; Koop–Korobilis; Heinrich–Reif; Stark–Croushore. These are the papers against which the gap should actually be stated. [75]

The most important change implied by this research is therefore **not** a change to the paper's hypotheses. The existing H1 and H2 definitions are, in fact, more defensible after the search. The change is in how the contribution must be described. H1 is grounded in a mature real-time information-flow literature; H2 is grounded in mature conditional/adaptive forecast-selection literature; and the paper's strongest opportunity is to make their distinction operational in a **controlled, pre-specified forecasting experiment**:

$$\boxed{\Delta^{I}_{k,t,s}(m) = L\left(e^{(m)}_{k,t,s+1}\right) - L\left(e^{(m)}_{k,t,s}\right)}$$

versus

$$\boxed{\Delta^{P}_{k,t,s} = L\left(e^{(m^{*}_{k,s})}_{k,t,s}\right) - L\left(e^{(m^{F}_{k})}_{k,t,s}\right)}.$$

The first asks **what additional information did** while holding the forecasting rule fixed. The second asks **what stage-conditioned model choice did** while holding the information set fixed. That separation is sharper than the generic statement that "later nowcasts improve" and avoids attributing to model adaptation gains that may simply arise from receiving more data. It is also narrow enough to respect the substantial prior art on release-by-release nowcasting, stage-varying specifications, model combinations, conditional selection and genuine real-time evaluation. [76]

For the eventual manuscript, the safest one-sentence description of the gap is therefore:

> **The contribution is not stage-dependent forecasting per se, but a pseudo-real-time design that separately identifies the predictive contribution of sequential information arrival and the incremental performance of a pre-specified release-stage model policy, evaluates the latter on observations excluded from policy selection, and assesses both effects under real-time and latest-vintage data across heterogeneous macroeconomic targets.**

That is the research position best supported by the literature reviewed as of August 2026.

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